

CHITOSE AP, Japan

WMO: 474340

Lat: 42.794N

Lon: 141.666E

Elev: 87

StdP: 14.65

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-0.3	3.6	-5.5	4.1	1.2	-0.7	5.3	6.3	28.9	31.2	25.5	30.1	2.7	210	0.626

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.1	82.3	72.6	79.0	71.0	76.8	69.8	74.9	79.8	73.1	77.0	71.6	74.9	10.9	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.4	124.7	77.8	71.8	117.7	75.5	70.1	110.9	73.6	38.5	79.4	36.8	76.5	35.5	75.0	80.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 25.3	22.0	19.3	DB	-6.4	87.1	4.3	2.9	-9.6	89.2	-12.1	90.9	-14.5	92.5	-17.7	94.6	(4)
(5)			WB	-7.0	77.2	4.3	1.9	-10.1	78.6	-12.6	79.7	-15.0	80.7	-18.1	82.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	45.1	19.8	21.1	30.0	42.1	52.4	59.8	67.3	70.0	63.3	50.7	38.1	25.1
	DBStd	18.45	6.77	7.09	6.50	5.78	5.19	4.82	4.77	4.76	5.79	6.06	7.37	7.33
	HDD50	3848	937	809	621	246	34	1	0	0	0	67	362	771
	HDD65	7587	1402	1229	1086	688	389	166	31	11	99	444	806	1236
	CDD50	2062	0	0	0	9	110	295	535	621	399	88	5	0
	CDD65	326	0	0	0	0	0	10	101	167	48	0	0	0
	CDH74	1218	0	0	0	0	10	45	355	647	161	0	0	0
	CDH80	198	0	0	0	0	1	5	53	126	13	0	0	0
Wind	WSAvg	8.7	7.1	7.9	9.2	10.6	11.1	9.9	9.5	8.9	8.3	7.9	7.7	6.8
Precipitation	PrecAvg	48.5	3.0	2.5	2.2	2.9	4.3	3.6	5.7	6.5	6.5	4.1	3.9	3.5
	PrecMax	61.8	4.3	5.6	4.0	5.2	7.4	10.0	11.9	13.3	10.9	5.9	8.1	6.6
	PrecMin	39.0	1.8	1.0	0.9	0.3	1.6	1.3	2.7	2.9	1.7	1.7	2.4	2.2
	PrecStd	5.6	0.6	1.0	0.8	1.4	1.5	1.7	2.1	2.7	2.6	1.1	1.3	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.4	41.2	51.6	66.6	74.8	78.9	84.5	87.8	82.0	70.2	61.1	49.8
		MCWB	36.8	37.0	43.5	53.0	58.5	66.1	74.2	76.5	72.1	60.9	54.9	47.4
	2%	DB	35.9	37.8	46.1	60.7	69.7	73.8	80.9	83.8	78.5	67.6	57.4	44.4
		MCWB	33.5	34.7	40.2	49.4	56.9	64.1	72.3	74.0	69.7	59.8	52.5	41.0
	5%	DB	33.9	35.9	42.6	56.8	66.2	71.5	78.4	80.4	75.3	64.5	54.0	39.5
		MCWB	31.7	33.1	38.3	47.6	54.7	62.8	70.8	72.6	68.2	57.5	49.7	36.3
	10%	DB	31.9	33.9	39.6	52.2	62.7	68.2	75.1	77.2	73.2	62.3	51.5	36.0
		MCWB	29.7	31.3	36.0	44.7	53.4	61.4	69.1	70.9	66.8	56.8	47.3	33.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	36.9	38.3	44.3	54.1	61.3	68.4	76.3	78.0	74.7	64.5	57.1	47.6
		MCDB	38.3	39.5	48.3	62.7	70.2	75.1	82.3	85.2	79.0	68.0	59.7	49.1
	2%	WB	34.0	35.4	41.3	51.1	58.7	65.8	74.0	75.6	72.6	61.5	53.9	41.2
		MCDB	35.4	37.1	44.5	57.7	65.9	71.8	79.0	81.0	75.9	65.2	56.9	43.3
	5%	WB	32.0	33.3	38.6	48.9	56.8	63.9	72.0	74.0	70.6	59.5	50.9	36.7
		MCDB	33.3	35.3	41.1	54.2	63.1	69.3	76.1	78.2	73.4	63.2	53.7	39.1
	10%	WB	29.2	31.4	36.5	46.4	55.1	62.3	70.2	72.5	68.3	57.2	47.8	33.5
		MCDB	31.3	33.7	39.3	50.8	60.6	66.9	73.7	75.9	71.6	61.1	50.7	35.7
Mean Daily Temperature Range	5% DB	MDBR	18.2	18.8	15.3	17.6	17.3	14.2	12.1	12.1	16.2	19.3	16.7	16.7
		MCDBR	16.9	17.4	17.0	24.2	23.8	19.4	17.1	15.8	17.0	20.6	20.7	17.8
		MCWBR	15.4	15.3	12.9	14.5	12.7	10.6	9.3	8.2	9.8	14.0	16.7	15.2
	5% WB	MCDBR	16.8	16.7	15.8	20.8	19.1	16.4	14.2	13.1	12.9	17.3	19.2	17.5
		MCWBR	16.1	15.3	13.0	13.7	11.6	9.6	8.5	7.7	9.0	13.4	16.9	16.1
Clear-Sky Solar Irradiance	taub	0.340	0.392	0.466	0.526	0.545	0.526	0.512	0.480	0.421	0.407	0.388	0.356	
	taud	2.036	1.889	1.788	1.782	1.834	1.962	2.069	2.174	2.295	2.191	2.139	2.052	
	Ebn at Noon	238	243	240	237	237	243	245	248	256	240	221	219	
	Edh at Noon	35	49	61	67	65	57	51	44	37	36	32	32	
All-Sky Solar Radiation	RadAvg	483	737	1106	1452	1621	1574	1426	1352	1169	857	515	389	
	RadStd	31	55	78	148	140	131	156	102	105	48	45	33	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (18 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-193	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page